1) a)

$h = \frac{2-0}{2} = 1$

	0	1	2
e^{-x^3}	1	e^{-1}	e^{-8}

$\int_0^2 e^{-x^3}\,dx \approx \frac{1}{3} \times 1 \times \{1 + e^{-8} + 4(e^{-1})\} = 0.823951$ (6 DP)

1) b)

$h = \frac{2-0}{4} = 0.5$

	0	0.5	1	1.5	2
e^{-x^3}	1	$e^{-0.125}$	e^{-1}	$e^{-3.375}$	e^{-8}

$\int_0^2 e^{-x^3}\,dx \approx \frac{1}{3} \times 0.5 \times \{1 + e^{-8} + 4(e^{-0.125} + e^{-3.375}) + 2(e^{-1})\} = 0.900492$ (6 DP)

1) c)

$h = \frac{2-0}{6} = \frac{1}{3}$

	0	$\frac{1}{3}$	$\frac{2}{3}$	1	$\frac{4}{3}$	$\frac{5}{3}$	2
e^{-x^3}	1	$e^{-\frac{1}{27}}$	$e^{-\frac{8}{27}}$	e^{-1}	$e^{-\frac{64}{27}}$	$e^{-\frac{125}{27}}$	e^{-8}

$\int_0^2 e^{-x^3}\,dx \approx \frac{1}{3} \times \frac{1}{3} \times \left\{1 + e^{-8} + 4\left(e^{-\frac{1}{27}} + e^{-1} + e^{-\frac{125}{27}}\right) + 2\left(e^{-\frac{8}{27}} + e^{-\frac{64}{27}}\right)\right\}$

$= 0.893275$ (6 DP)

2) a)

$h = \frac{2-1}{2} = 0.5$

	1	1.5	2
$\frac{1}{1+\ln x}$	1	$\frac{1}{1+\ln 1.5}$	$\frac{1}{1+\ln 2}$

$\int_1^2 \frac{1}{1+\ln x}\,dx \approx \frac{1}{3} \times 0.5 \times \left\{1 + \frac{1}{1+\ln 2} + 4\left(\frac{1}{1+\ln 1.5}\right)\right\} = 0.739442$ (6 DP)

2) b)

$h = \frac{2-1}{4} = 0.25$

	1	1.25	1.5	1.75	2
$\frac{1}{1+\ln x}$	1	$\frac{1}{1+\ln 1.25}$	$\frac{1}{1+\ln 1.5}$	$\frac{1}{1+\ln 1.75}$	$\frac{1}{1+\ln 2}$

$\int_1^2 \frac{1}{1+\ln x}\ \mathrm{d}x \approx \frac{1}{3} \times 0.25 \times \left\{1 + \frac{1}{1+\ln 2} + 4\left(\frac{1}{1+\ln 1.25} + \frac{1}{1+\ln 1.75}\right) + 2\left(\frac{1}{1+\ln 1.5}\right)\right\}$

$= 0.737386$ (6 DP)

2) c)

$h = \frac{2-1}{6} = \frac{1}{6}$

	1	$\frac{7}{6}$	$\frac{4}{3}$	1.5	$\frac{5}{3}$	$\frac{11}{6}$	2
$\frac{1}{1+\ln x}$	1	$\frac{1}{1+\ln\frac{7}{6}}$	$\frac{1}{1+\ln\frac{4}{3}}$	$\frac{1}{1+\ln 1.5}$	$\frac{1}{1+\ln\frac{5}{3}}$	$\frac{1}{1+\ln\frac{11}{6}}$	$\frac{1}{1+\ln 2}$

$\int_1^2 \frac{1}{1+\ln x}\ \mathrm{d}x \approx \frac{1}{3} \times \frac{1}{6} \times \left\{1 + \frac{1}{1+\ln 2} + 4\left(\frac{1}{1+\ln\frac{7}{6}} + \frac{1}{1+\ln 1.5} + \frac{1}{1+\ln\frac{11}{6}}\right) + 2\left(\frac{1}{1+\ln\frac{4}{3}} + \frac{1}{1+\ln\frac{5}{3}}\right)\right\}$

$= 0.737212$ (6 DP)

3)

$\frac{1}{3} \times 2 \times \{0 + 15.2 + 4(5.6 + 13.0) + 2(9.8)\} = \frac{1}{3} \times 218.4 = 72.8$ metres

4)

$h = \frac{1-0}{4} = 0.25$

	0	0.25	0.5	0.75	1
$\cosh x$	1	$\cosh 0.25$	$\cosh 0.5$	$\cosh 0.75$	$\cosh 1$

$\int_0^1 \cosh x\ \mathrm{d}x \approx \frac{1}{3} \times 0.25 \times \{1 + \cosh 1 + 4(\cosh 0.25 + \cosh 0.75) + 2(\cosh 0.5)\}$

$= 1.18$ (3 SF)

5) a)

$h = \frac{0.5-0.1}{4} = 0.1$

	0.1	0.2	0.3	0.4	0.5
$\frac{\tan x}{x}$	$\frac{\tan 0.1}{0.1}$	$\frac{\tan 0.2}{0.2}$	$\frac{\tan 0.3}{0.3}$	$\frac{\tan 0.4}{0.4}$	$\frac{\tan 0.5}{0.5}$

$\int_{0.1}^{0.5} \frac{\tan x}{x}\,\mathrm{d}x \approx \frac{1}{3} \times 0.1 \times \left\{\frac{\tan 0.1}{0.1} + \frac{\tan 0.5}{0.5} + 4\left(\frac{\tan 0.2}{0.2} + \frac{\tan 0.4}{0.4}\right) + 2\left(\frac{\tan 0.3}{0.3}\right)\right\} = 0.415$ (3 SF)

5) b)

Increase the number of strips used.

6) a)

$h = \frac{\pi-0}{2} = \frac{\pi}{2}$

	0	$\frac{\pi}{2}$	π
$\mathrm{e}^{\cos x}$	e^{1}	1	e^{-1}

$\int_{1}^{2} \frac{1}{1+\ln x}\,\mathrm{d}x \approx \frac{1}{3} \times \frac{\pi}{2} \times \{\mathrm{e}^{1} + \mathrm{e}^{-1} + 4(1)\} = 3.7103$ (4 DP)

6) b)

Increase the number of ordinates used.

7) a)

$h = \frac{1-0}{4} = 0.25$

	0	0.25	0.5	0.75	1
e^{x}	1	$\mathrm{e}^{0.25}$	$\mathrm{e}^{0.5}$	$\mathrm{e}^{0.75}$	e^{1}

$\int_{0}^{1} \mathrm{e}^{x}\,\mathrm{d}x \approx \frac{1}{3} \times 0.25 \times \{1 + \mathrm{e}^{1} + 4(\mathrm{e}^{0.25} + \mathrm{e}^{0.75}) + 2(\mathrm{e}^{0.5})\} = 1.7183$ (5 SF)

7) b)

$\int_{0}^{1} \mathrm{e}^{x}\,\mathrm{d}x = [\mathrm{e}^{x}]_{0}^{1} = \mathrm{e} - 1$

7) c)

$\frac{1.7183-(\mathrm{e}-1)}{\mathrm{e}-1} = 0.000\,010\,6$ (3 SF)

8) a)

$h = \frac{1-0}{4} = 0.25$

	0	0.25	0.5	0.75	1
$x^2 + e^{2x}$	1	$0.0625 + e^{0.5}$	$0.25 + e^1$	$0.5625 + e^{1.5}$	$1 + e^2$

$\int_0^1 x^2 + e^{2x}\,dx$

$\approx \frac{1}{2} \times 0.25 \times \{1 + 1 + e^2 + 2(0.0625 + e^{0.5} + 0.25 + e^1 + 0.5625 + e^{1.5})\}$

$= 3.6046$ (4 DP)

8) b)

$\int_0^1 (x^2 + e^{2x})\,dx$

$\approx \frac{1}{3} \times 0.25 \times \{1 + 1 + e^2 + 4(0.0625 + e^{0.5} + 0.5625 + e^{1.5}) + 2(0.25 + e^1)\}$

$= 3.5289$ (4 DP)

8) c)

$\int_0^1 x^2 + e^{2x}\,dx = \left[\frac{1}{3}x^3 + \frac{1}{2}e^{2x}\right]_0^1 = \left(\frac{1}{3} + \frac{1}{2}e^2\right) - \left(\frac{1}{2}\right) = \frac{1}{2}e^2 - \frac{1}{6}$

9) a)

$h = \frac{2-0}{2} = 1$

	0	1	2
$x\sqrt{2x^2+1}$	0	$\sqrt{3}$	6

$\int_0^2 x\sqrt{2x^2+1}\,dx \approx \frac{1}{3} \times 1 \times \{0 + 6 + 4(\sqrt{3})\} = 4.3094$ (4 DP)

9) b)

$\int_0^2 x\sqrt{2x^2+1}\,dx = \int_0^2 x(2x^2+1)^{\frac{1}{2}}\,dx = \left[\frac{1}{6}(2x^2+1)^{\frac{3}{2}}\right]_0^2 = \frac{27}{6} - \frac{1}{6} = \frac{13}{3}$

9) c)

$\frac{4.3094 - \left(\frac{13}{3}\right)}{\frac{13}{3}} = -0.00552$

10) a)

$h = \frac{1-0}{4} = 0.25$

	0	0.25	0.5	0.75	1
$\cos(x^2)$	1	$\cos 0.0625$	$\cos 0.25$	$\cos 0.5625$	$\cos 1$

$\int_0^1 \cos(x^2)\ \mathrm{d}x \approx \frac{1}{3} \times 0.25 \times \{1 + \cos 1 + 4(\cos 0.0625 + \cos 0.5625) + 2(\cos 0.25)\}$

$= 0.9045$ (4 DP)

10) b)

$\mathrm{f}(x) = 5\cos((x-1)^2)$

10) c)

$\int_1^2 \mathrm{f}(x)\ \mathrm{d}x \approx 5 \times 0.9045 = 4.523$ (3 DP)

10) d)

	1	1.25	1.5	1.75	2
$5\cos((x-1)^2)$	5	$5\cos 0.0625$	$5\cos 0.25$	$5\cos 0.5625$	$5\cos 1$

$\int_1^2 \mathrm{f}(x)\ \mathrm{d}x \approx \frac{1}{3} \times 0.25 \times \{5 + 5\cos 1 + 4(5\cos 0.0625 + 5\cos 0.5625) + 2(5\cos 0.25)\}$

$= 4.523$ (3 DP)

11) a)

$h = \frac{0.3-0.1}{4} = 0.05$

	0.1	0.15	0.2	0.25	0.3
$\frac{1}{\sqrt{1-x^3}}$	$\frac{1}{\sqrt{1-0.1^3}}$	$\frac{1}{\sqrt{1-0.15^3}}$	$\frac{1}{\sqrt{1-0.2^3}}$	$\frac{1}{\sqrt{1-0.25^3}}$	$\frac{1}{\sqrt{1-0.3^3}}$

$\int_{0.1}^{0.3} \frac{1}{\sqrt{1-x^3}}\ \mathrm{d}x \approx \frac{1}{3} \times 0.05 \times \left\{\frac{1}{\sqrt{1-0.1^3}} + \frac{1}{\sqrt{1-0.3^3}} + 4\left(\frac{1}{\sqrt{1-0.15^3}} + \frac{1}{\sqrt{1-0.25^3}}\right) + 2\left(\frac{1}{\sqrt{1-0.2^3}}\right)\right\}$

$= 0.201$ (3 DP)

11) b)

$\int_{0.1}^{0.3} \frac{5}{\sqrt{1-x^3}}\ \mathrm{d}x \approx 1.005$ (3 DP)

12) a)

$h = \frac{3-1}{4} = 0.5$

	1	1.5	2	2.5	3
$\frac{\sin x}{x}$	$\sin 1$	$\frac{\sin 1.5}{1.5}$	$\frac{\sin 2}{2}$	$\frac{\sin 2.5}{2.5}$	$\frac{\sin 3}{3}$

$\int_1^3 \frac{\sin x}{x}\,\mathrm{d}x \approx \frac{1}{3} \times 0.5 \times \left\{\sin 1 + \frac{\sin 3}{3} + 4\left(\frac{\sin 1.5}{1.5} + \frac{\sin 2.5}{2.5}\right) + 2\left(\frac{\sin 2}{2}\right)\right\}$

$= 0.903$ (3 DP)

12) b)

$\int_1^5 \left(3 - \frac{\sin x}{x}\right)\,\mathrm{d}x \approx 2 \times 3 - 0.903 = 5.097$

13) a)

$\pi \int_0^2 x^2(\ln(x+2))^2\,\mathrm{d}x$

13) b)

$h = \frac{2-0}{4} = 0.5$

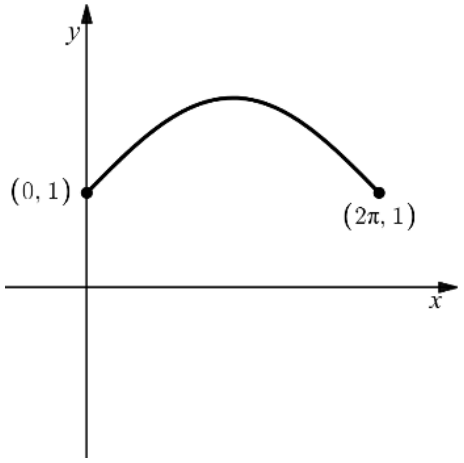
	0	0.5	1	1.5	2
$x^2(\ln(x+2))^2$	0	$0.25(\ln 2.5)^2$	$(\ln 3)^2$	$2.25(\ln 3.5)^2$	$4(\ln 4)^2$

$\pi \int_0^2 x^2(\ln(x+2))^2\,\mathrm{d}x$

$\approx \pi \times \frac{1}{3} \times 0.5 \times \{0 + 4(\ln 4)^2 + 4(0.25(\ln 2.5)^2 + 2.25(\ln 3.5)^2) + 2((\ln 3)^2)\}$

$= 13.1$ (3 SF)

14) a)



14) b)

$h = \frac{2\pi - 0}{6} = \frac{\pi}{3}$

	0	$\frac{\pi}{3}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{5\pi}{3}$	2π
$1 + \sin\frac{x}{2}$	1	$1 + \sin\frac{\pi}{6}$	$1 + \sin\frac{\pi}{3}$	$1 + \sin\frac{\pi}{2}$	$1 + \sin\frac{2\pi}{3}$	$1 + \sin\frac{5\pi}{6}$	$1 + \sin\pi$

$\int_0^{2\pi} \left(1 + \sin\frac{x}{2}\right) \mathrm{d}x$

$\approx \frac{1}{3} \times \frac{\pi}{3} \times \left\{2 + \sin\pi + 4\left(3 + \sin\frac{\pi}{6} + \sin\frac{\pi}{2} + \sin\frac{5\pi}{6}\right) + 2\left(2 + \sin\frac{\pi}{3} + \sin\frac{2\pi}{3}\right)\right\}$

$= 10.284912$ (6 DP)

14) c)

$\frac{1}{2\pi}\int_0^{2\pi} \left(1 + \sin\frac{x}{2}\right) \mathrm{d}x \approx 1.63689$ (5 DP)

15)

$h = \frac{7-1}{6} = 1$

	1	2	3	4	5	6	7
$\frac{1}{x}$	1	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{5}$	$\frac{1}{6}$	$\frac{1}{7}$

$\int_1^7 \frac{1}{x} \mathrm{d}x \approx \frac{1}{3} \times 1 \times \left\{1 + \frac{1}{7} + 4\left(\frac{1}{2} + \frac{1}{4} + \frac{1}{6}\right) + 2\left(\frac{1}{3} + \frac{1}{5}\right)\right\} = 1.9587$ (4 DP)

16) a)

$\sin 2x \approx 2x - \frac{4x^3}{3} + \frac{4x^5}{15}$

16) b)

$h = \frac{1-0}{4} = 0.25$

	0	0.25	0.5	0.75	1
$2x - \frac{4x^3}{3} + \frac{4x^5}{15}$	0	$\frac{1841}{3840}$	$\frac{101}{120}$	$\frac{1281}{1280}$	$\frac{14}{15}$

$\int_0^1 2x - \frac{4x^3}{3} + \frac{4x^5}{15} \mathrm{d}x \approx \frac{1}{3} \times 0.25 \times \left\{0 + \frac{14}{15} + 4\left(\frac{1841}{3840} + \frac{1281}{1280}\right) + 2\left(\frac{101}{120}\right)\right\} = 0.7115$ (4 DP)

17) a)

$x^2 \ln(1+x) \approx x^2\left(x - \frac{x^2}{2} + \frac{x^3}{3}\right) = x^3 - \frac{x^4}{2} + \frac{x^5}{3}$

17) b)

$h = \frac{0.4-0}{4} = 0.1$

	0	0.1	0.2	0.3	0.4
$x^3 - \frac{x^4}{2} + \frac{x^5}{3}$	0	$\frac{143}{150000}$	$\frac{137}{18750}$	$\frac{297}{12500}$	$\frac{512}{9375}$

$\int_0^{0.4}\left(x^3 - \frac{x^4}{2} + \frac{x^5}{3}\right)\,\mathrm{d}x \approx \frac{1}{3} \times 0.1 \times \left\{0 + \frac{512}{9375} + 4\left(\frac{143}{150000} + \frac{297}{12500}\right) + 2\left(\frac{137}{18750}\right)\right\}$

$= 0.0056$ (4 DP)

18) a)

$y = -1^1 + 4 = 3$ so $(1, 3)$

18) b)

$h = \frac{2-1}{4} = 0.25$

	1	1.25	1.5	1.75	2
$-x^x + 4$	3	2.67829	2.16288	1.33733	0

$\int_0^2(-x^x + 4)\,\mathrm{d}x \approx \frac{1}{3} \times 0.25 \times \{3 + 0 + 4(2.67829 + 1.33733) + 2(2.16288)\}$

$= 1.95$ (2 DP)

18) c)

$\frac{\mathrm{d}}{\mathrm{d}x}(x^x) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\mathrm{e}^{\ln(x^x)}\right) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\mathrm{e}^{x\ln x}\right)$

So that

$\frac{\mathrm{d}}{\mathrm{d}x}\left(\mathrm{e}^{x\ln x}\right) = \frac{\mathrm{d}}{\mathrm{d}x}(x\ln x) \times \mathrm{e}^{x\ln x}$

$= \left(x \times \frac{1}{x} + \ln x\right) \times \mathrm{e}^{x\ln x}$

$= (1 + \ln x) \times \mathrm{e}^{\ln x^x}$

$= x^x(1 + \ln x)$ as req.

18) d)

When $x = 1, y = 3$

If $y = -x^x + 4$ then $\frac{\mathrm{d}y}{\mathrm{d}x} = -x^x(1 + \ln x)$

$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=1} = -1^1(1 + \ln 1) = -1$

$y - 3 = -1(x - 1)$

$y + x = 4$

18) e)

$P(1, 3)$ and $(4,0)$

Shaded area:

$\frac{1}{2} \times 3 \times 3 - 1.95 = 2.55$ (2 DP)

19) a) i)

$h = \frac{2-0}{4} = 0.5$

	0	0.5	1	1.5	2
x^2	0	0.25	1	2.25	4

$\int_0^2 x^2 \, \mathrm{d}x \approx \frac{1}{2} \times 0.5 \times \{0 + 4 + 2(0.25 + 1 + 2.25)\} = 2.75$

19) a) ii)

	0		0.5		1		1.5		2
$x_{i-\frac{1}{2}}$		0.25		0.75		1.25		1.75	
x^2		0.0625		0.25		1.5625		3.0625	

$\int_0^2 x^2 \, \mathrm{d}x \approx 0.5 \times \{0.0625 + 0.5625 + 1.5625 + 3.0625\} = 2.625$

19) a) iii)

	0	0.5	1	1.5	2
x^2	0	0.25	1	2.25	4

$\int_0^2 x^2 \, \mathrm{d}x \approx \frac{1}{3} \times 0.5 \times \{0 + 4 + 4(0.25 + 2.25) + 2(1)\} = \frac{8}{3}$ or $2.\dot{6}$

19) b)

$\int_0^2 x^2\,\mathrm{d}x = \left[\frac{1}{3}x^3\right]_0^2 = \frac{1}{3}\times 2^3 - 0 = \frac{8}{3}$

19) c)

$\frac{2\times 2.625+2.75}{3} = \frac{8}{3} = S_4$ as req.

20) a) i)

$h = \frac{6-0}{4} = 1.5$

	0	1.5	3	4.5	6
$\sqrt{x+1}$	1	$\sqrt{2.5}$	2	$\sqrt{5.5}$	$\sqrt{7}$

$\int_0^6 \sqrt{x+1}\,\mathrm{d}x \approx \frac{1}{2}\times 1.5\times\left\{1+\sqrt{7}+2\left(\sqrt{2.5}+2+\sqrt{5.5}\right)\right\} = 11.623834$ (6 DP)

20) a) ii)

	0		1.5		3		4.5		6
$x_{i-\frac{1}{2}}$		0.75		2.25		3.75		5.25	
$\sqrt{x+1}$		$\sqrt{1.75}$		$\sqrt{3.25}$		$\sqrt{4.75}$		$\sqrt{6.25}$	

$\int_0^6 \sqrt{x+1}\,\mathrm{d}x \approx 1.5\times\left\{\sqrt{1.75}+\sqrt{3.25}+\sqrt{4.75}+\sqrt{6.25}\right\} = 11.707651$ (6 DP)

20) a) iii)

	0	0.75	1.5	2.25	3
$\sqrt{x+1}$	1	$\sqrt{1.75}$	$\sqrt{2.5}$	$\sqrt{3.25}$	2
	3.75	4.5	5.25	6	
$\sqrt{x+1}$	$\sqrt{4.75}$	$\sqrt{5.5}$	$\sqrt{6.25}$	$\sqrt{7}$	

$\int_0^6 \sqrt{x+1}\,\mathrm{d}x$

$\approx \frac{1}{3}\times 0.75\times\left\{1+\sqrt{7}+4\left(\sqrt{1.75}+\sqrt{3.25}+\sqrt{4.75}+\sqrt{6.25}\right)+2\left(\sqrt{2.5}+2+\sqrt{5.5}\right)\right\}$

$= 11.679712$ (6 DP)

20) b)

$\int_0^6 \sqrt{x+1}\,\mathrm{d}x = \int_0^6 (x+1)^{\frac{1}{2}}\,\mathrm{d}x = \left[\frac{2}{3}(x+1)^{\frac{3}{2}}\right]_0^6 = \frac{2}{3}\times 7^{\frac{3}{2}} - \frac{2}{3} = \frac{2}{3}\left(7\sqrt{7}-1\right)$

20) c)

$\frac{2\times 11.707651 + 11.623834}{3} = 11.679712 = S_4$ as req.

21) a)

$h = \frac{b-a}{2}$

	a	$\frac{a+b}{2}$	b
x^2	a^2	$\frac{(a+b)^2}{4}$	b^2

$\int_a^b x^2\,\mathrm{d}x \approx \frac{1}{2}\times\frac{a+b}{2}\times\left\{a^2+b^2+2\left(\frac{(a+b)^2}{4}\right)\right\} = \frac{1}{4}(b-a)\left(a^2+\frac{1}{2}(a+b)^2+b^2\right)$

$= \frac{1}{8}(b-a)(2a^2+(a+b)^2+2b^2)$

$= \frac{1}{8}(b-a)(3a^2+2ab+3b^2)$

21) b)

	a		$\frac{a+b}{2}$		b
$x_{i-\frac{1}{2}}$		$\frac{1}{4}(3a+b)$		$\frac{1}{4}(a+3b)$	
x^2		$\frac{1}{16}(3a+b)^2$		$\frac{1}{16}(a+3b)^2$	

$\int_a^b x^2\,\mathrm{d}x \approx \frac{b-a}{2}\times\left\{\frac{1}{16}(3a+b)^2+\frac{1}{16}(a+3b)^2\right\} = \frac{1}{32}(b-a)\{(3a+b)^2+(a+3b)^2\}$

$= \frac{1}{32}(b-a)\{9a^2+6ab+b^2+a^2+6ab+9b^2\}$

$= \frac{1}{32}(b-a)\{10a^2+12ab+10b^2\}$

$= \frac{1}{16}(b-a)(5a^2+6ab+5b^2)$

21) c)

	a	$\frac{3a+b}{4}$	$\frac{a+b}{2}$	$\frac{a+3b}{4}$	b
x^2	a^2	$\frac{(3a+b)^2}{16}$	$\frac{(a+b)^2}{4}$	$\frac{(a+3b)^2}{16}$	b^2

$\int_a^b x^2\,\mathrm{d}x \approx \frac{1}{3}\times\frac{b-a}{4}\times\left\{a^2+b^2+4\left(\frac{(3a+b)^2}{16}+\frac{(a+3b)^2}{16}\right)+2\left(\frac{(a+b)^2}{4}\right)\right\}$

$=\frac{1}{3}\times\frac{b-a}{4}\times\left\{a^2+b^2+\frac{1}{4}(9a^2+6ab+b^2+a^2+6ab+9b^2)+\frac{1}{2}(a^2+2ab+b^2)\right\}$

$=\frac{1}{3}\times\frac{b-a}{4}\times\left\{a^2+b^2+\frac{1}{4}(10a^2+12ab+10b^2)+\frac{1}{2}(a^2+2ab+b^2)\right\}$

$=\frac{1}{3}\times\frac{b-a}{4}\times(4a^2+4b^2+4ab)$

$=\frac{1}{3}(b-a)(a^2+ab+b^2)$

21) d)

Since $\frac{1}{3}(b-a)(a^2+ab+b^2) = \frac{1}{3}(a^2b+ab^2+b^3-a^3-a^2b-ab^2) = \frac{1}{3}(b^3-a^3)$

$\int_a^b x^2\,\mathrm{d}x = \left[\frac{1}{3}x^3\right]_a^b = \frac{1}{3}(b^3-a^3) = \frac{1}{3}(b-a)(a^2+ab+b^2)$

21) e)

$2M_2+T_2 = 2\times\frac{1}{16}(b-a)\{5a^2+6ab+5b^2\}+\frac{1}{8}(b-a)(3a^2+2ab+3b^2)$

$= \frac{1}{8}(b-a)\{5a^2+6ab+5b^2+3a^2+2ab+3b^2\}$

$= \frac{1}{8}(b-a)(8a^2+8ab+8b^2)$

$= (b-a)(a^2+ab+b^2)$

So that

$\frac{2M_2+T_2}{3} = \frac{1}{3}(b-a)(a^2+ab+b^2) = S_4$ as req.